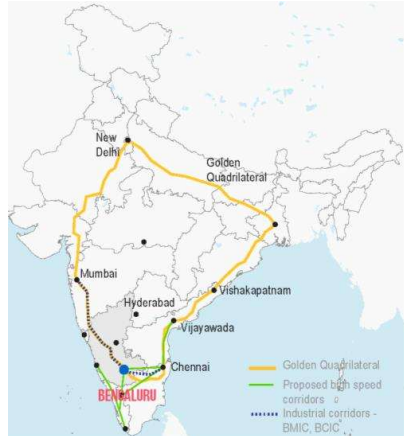


SMART CITY PROPOSAL, **Bengaluru**

CITY PROFILE

City context & Demographics



POPULATION
9.8 Million (2015)

AREA OF CITY – 709.5 sq km
198 wards

POPULATION DENSITY
4378 persons/ sq km

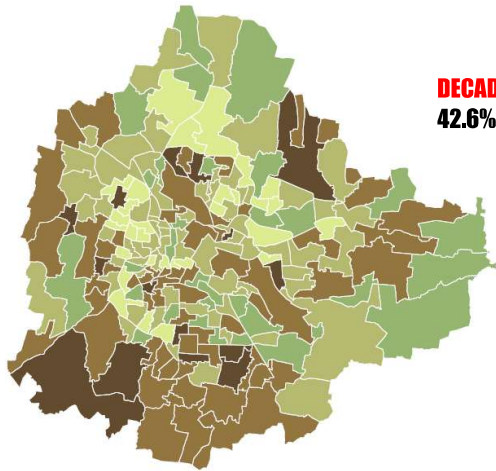


Literate males
3,595,059

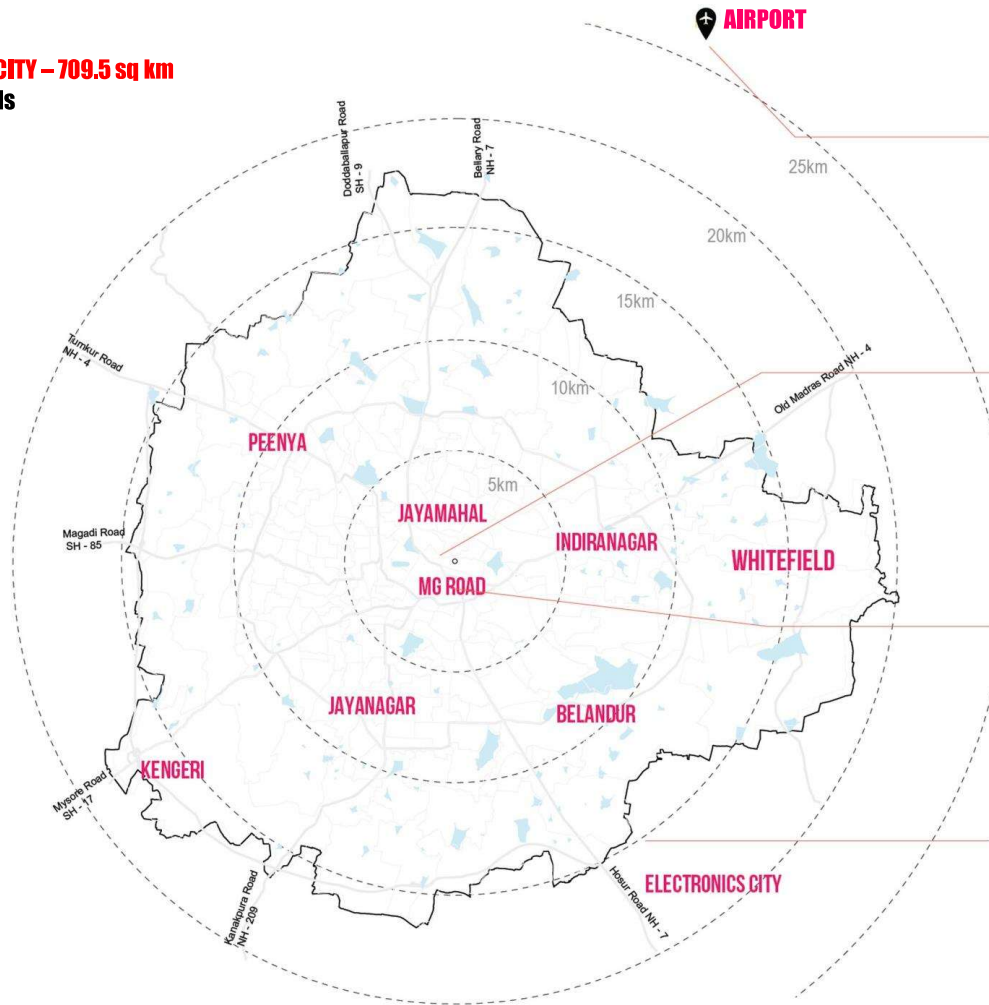
Literate females
3,082,135

**Sex ratio – 923 females
for every 1000 males**

DECADAL GROWTH RATE
42.6% Pan City



<0% 0.1-25 26-50 50-150 151-315%



Kempegowda International Airport



Vidhana Soudha - State legislature building



Commercial Street and Brigade road



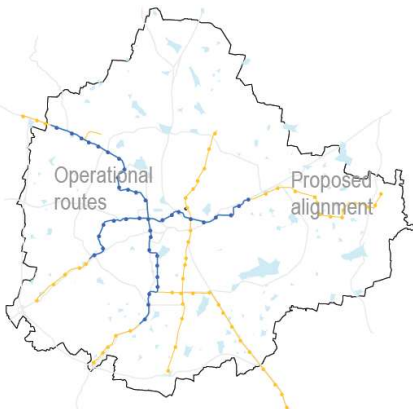
Electronics city - IT industry



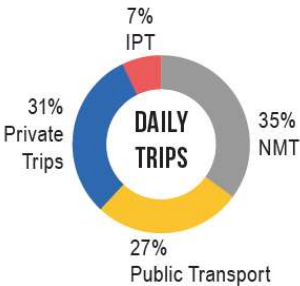
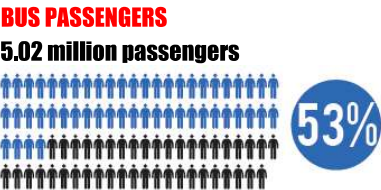
SMART BENGALURU

CITY PROFILE

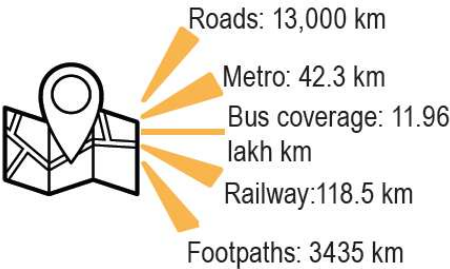
Transportation & Connectivity



METRO NETWORK
42.3 km operational, 72.09 km proposed



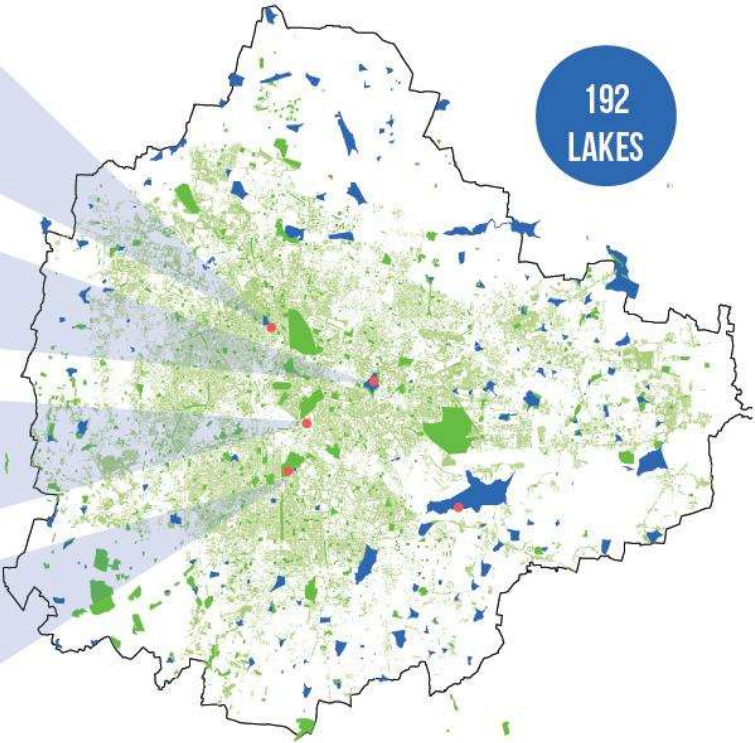
Source: mobility indicators, DULT -2010



Environment



LAKES & GREEN SPACES
17.5 sq km & 27.9 sq km



AREAS OF CONCERN

Severe traffic congestion

Inadequate public transport facilities

Inadequate pedestrian and disable friendly infrastructure

Reduction of open spaces/ lakes

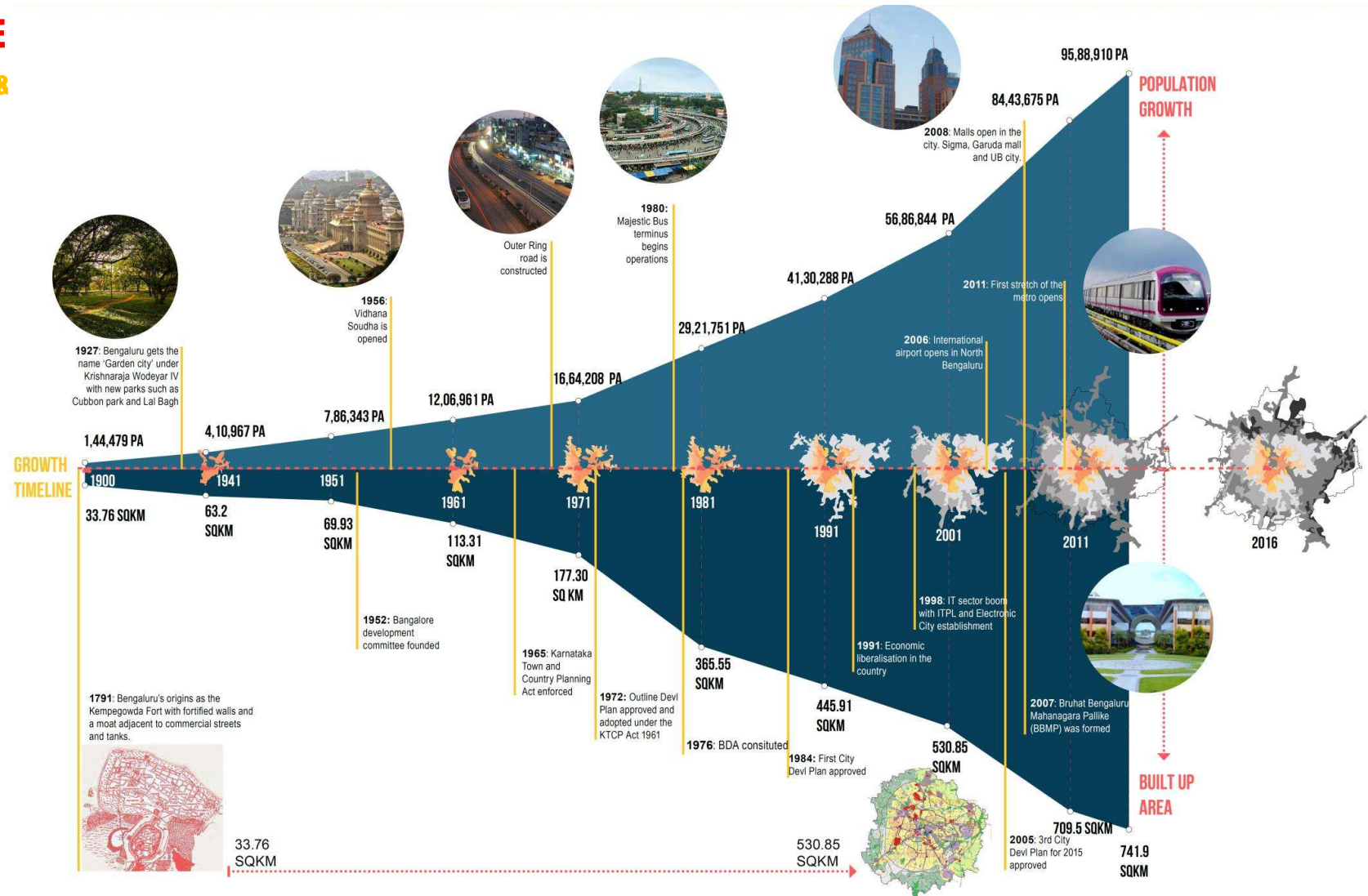
Contamination of water bodies

Sub optimal solid waste management



CITY PROFILE

Evolution &



CITY PROFILE

Self Assessment



MOBILITY AND TRANSPORT

13000 km road network

2428 kms (18%) of public bus transit system

52L (53%) Daily ridership in BMTC

42.30 kms of Metro rail network

1% signalised intersection

26% footpath available



WATER SUPPLY

94 LPCD Water supply

38% coverage of water supply connection

50% quality of water supply

49% Non revenue water

95.5% metered connection
1400 smart meters

2% Houses with rain water harvesting facility



WASTE WATER & SWM

66.3% Coverage of Sewerage network

62.5% sewage treated in STPs

30 - 40% Segregation at source

60% collection at source



STORM WATER DRAINAGE

850 km Storm water drain

47% natural stream missing

38% Underground drainage network



SAFETY AND SECURITY

50% reduction in road accidents

21758 repeated offenders driving licenses sent for suspension

400 surveillance cameras

Source: **Bangalore Traffic Police**

Centre for Sustainable Development

BBMP

Street Quality Score

CITIZEN PARTICIPATION

15,20,414
RESPONSES

52 LAKH+ CITIZENS CONTACTED

TOTAL CAMPAIGN REACH

10.37 LAKH+ CITIZENS VOTED

CITY LEVEL VOTES

3 LAKH+ CITIZEN INPUTS

NEIGHBOURHOOD INPUTS

68,494 CITIZEN OPINIONS

SPECIFIC OPINIONS



ಬೃಹತ್ ಬೆಂಗಳೂರು ಮಹಾನಗರ ಪಾಲಿಕೆ
Bruhat Bengaluru Mahanagara Palike

Smart City
Namma
Bengaluru
Vote Now!



LIKES ON FB
6,24,891



TWEETS 5681



WHATSAPP MSGS
25,435



49,229 LIKES ON
SOCIAL MEDIA



2,52,422 DIRECT
BALLOTS



1,12,433 MISSED
CALL VOTES



50 LAKH SMS SENT



YOUTUBE VIDEOS



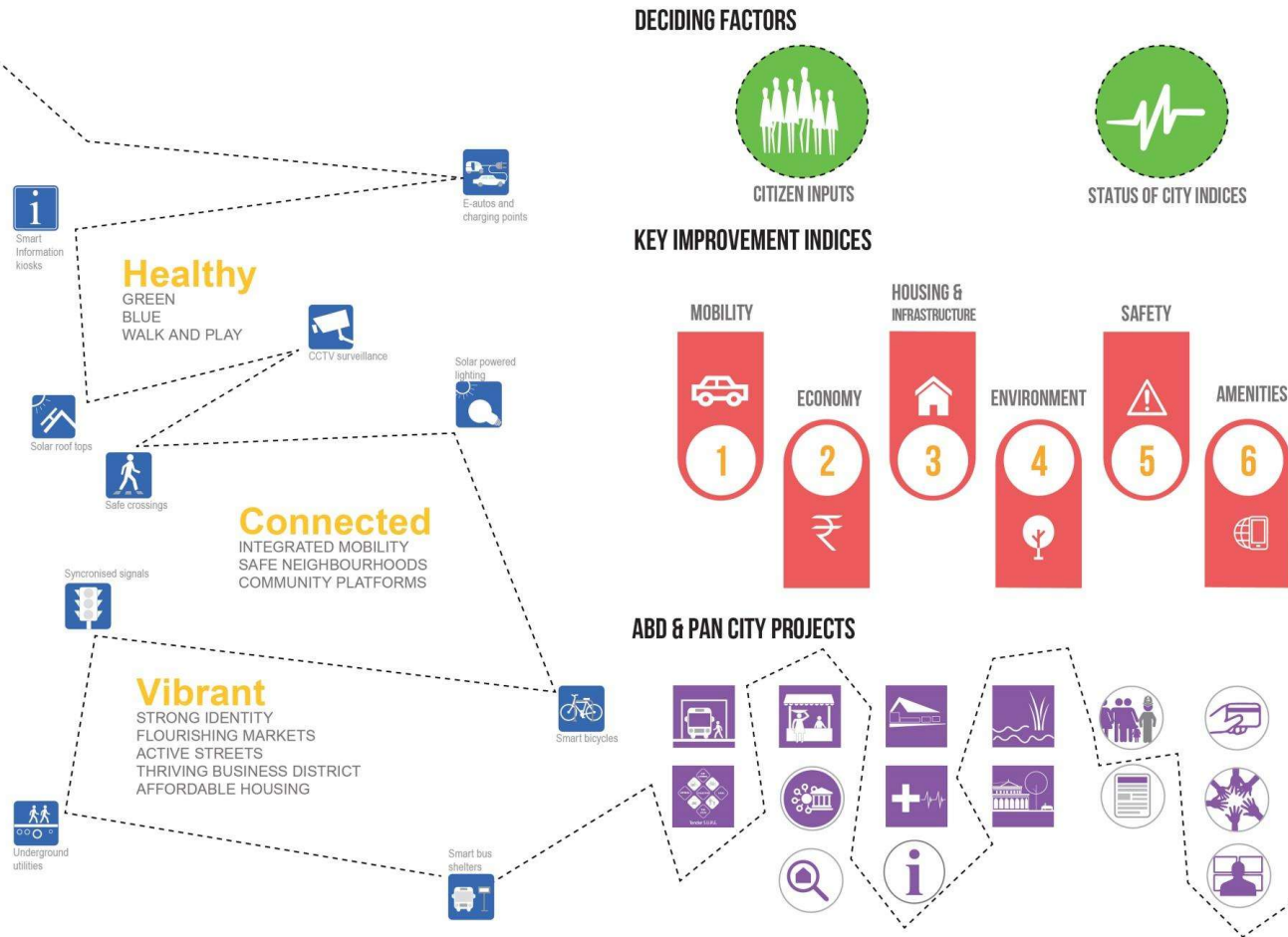
67,114 INPUTS ON THE
BUDGET PRIORITIES



291 INPUTS ON
MYGOV.IN

VISION FOR BENGALURU

Livable Bengaluru



...And, a smart city is built by building smart communities



PROPOSED INTERVENTIONS

ABD & Pan City



7

AREA BASED DEVELOPMENT PROJECTS

10 areas of interventions

PROJECT 1: **Revitalization of historic heart of city**

- Road redevelopment under Tender S.U.R.E. model

PROJECT 2: **Integrated mobility towards creating vibrant destination**

- Shivajinagar bus station and Russell Market
- Kempegowda Bus Stand (Majestic)

PROJECT 3: **Redevelopment of historic economic centres**

- KR Market
- Malleshwaram Market

PROJECT 4: **Innovation of downstream clean up of drainage system**

- Ulsoor lake
- Sankey Tank

PROJECT 5: **Protection & redevelopment of centrally located parkland**

- Cubbon Park

PROJECT 6: **Increasing affordable housing stock through slum redevelopment**

- Swathanthra Palya Slum redevelopment, Gandhinagar

PROJECT 7: **Retrofitting of a health care facility**

- K.C General Hospital

1

PAN CITY PROJECT

8 proposals on governance and public services

PROJECT 1: **CITIZEN CENTRIC E-GOVERNANCE AND SERVICES**

GOVERNANCE

- Municipal finance reforms
- Participatory budgeting
- Improvement in property tax collections
- Grievance management
- Online project information system for public project management
- Improvement of neighborhood safety

PUBLIC SERVICES

- BTRIPS – Bengaluru Travel Related Information and Planning System
- Open data portal for citizen information and innovation

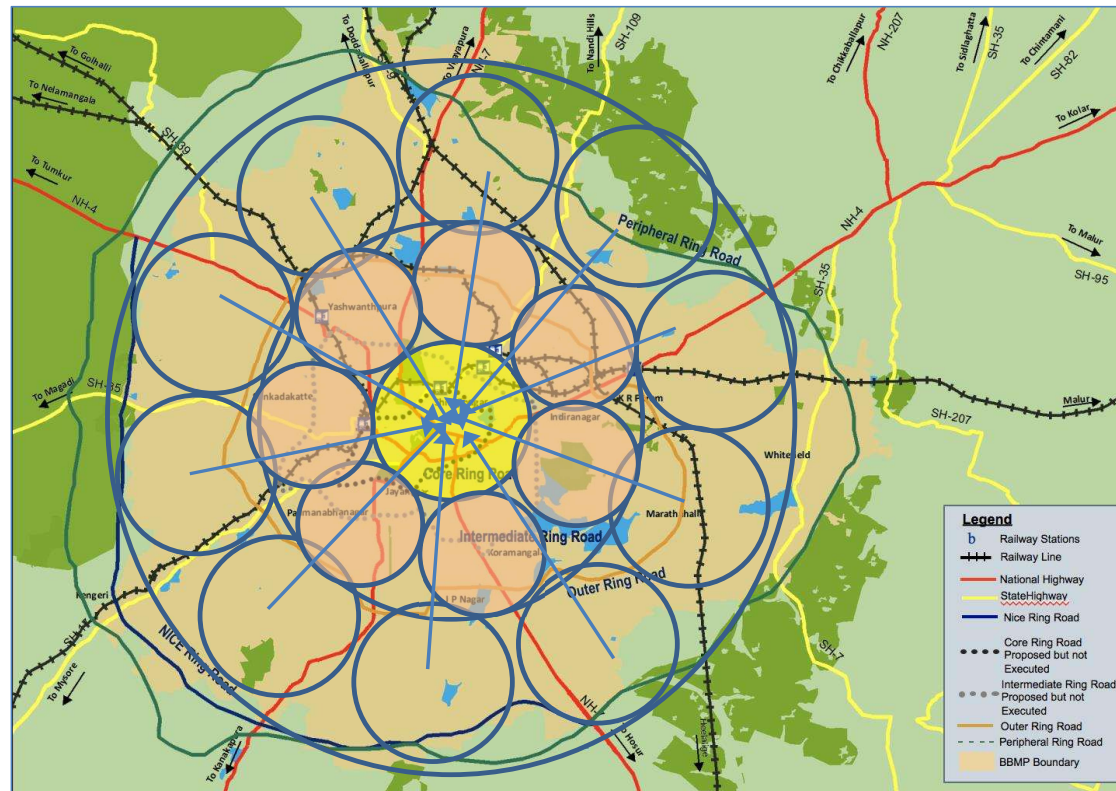
AREA BASED DEVELOPMENT

Revitalizing the city center of Bengaluru



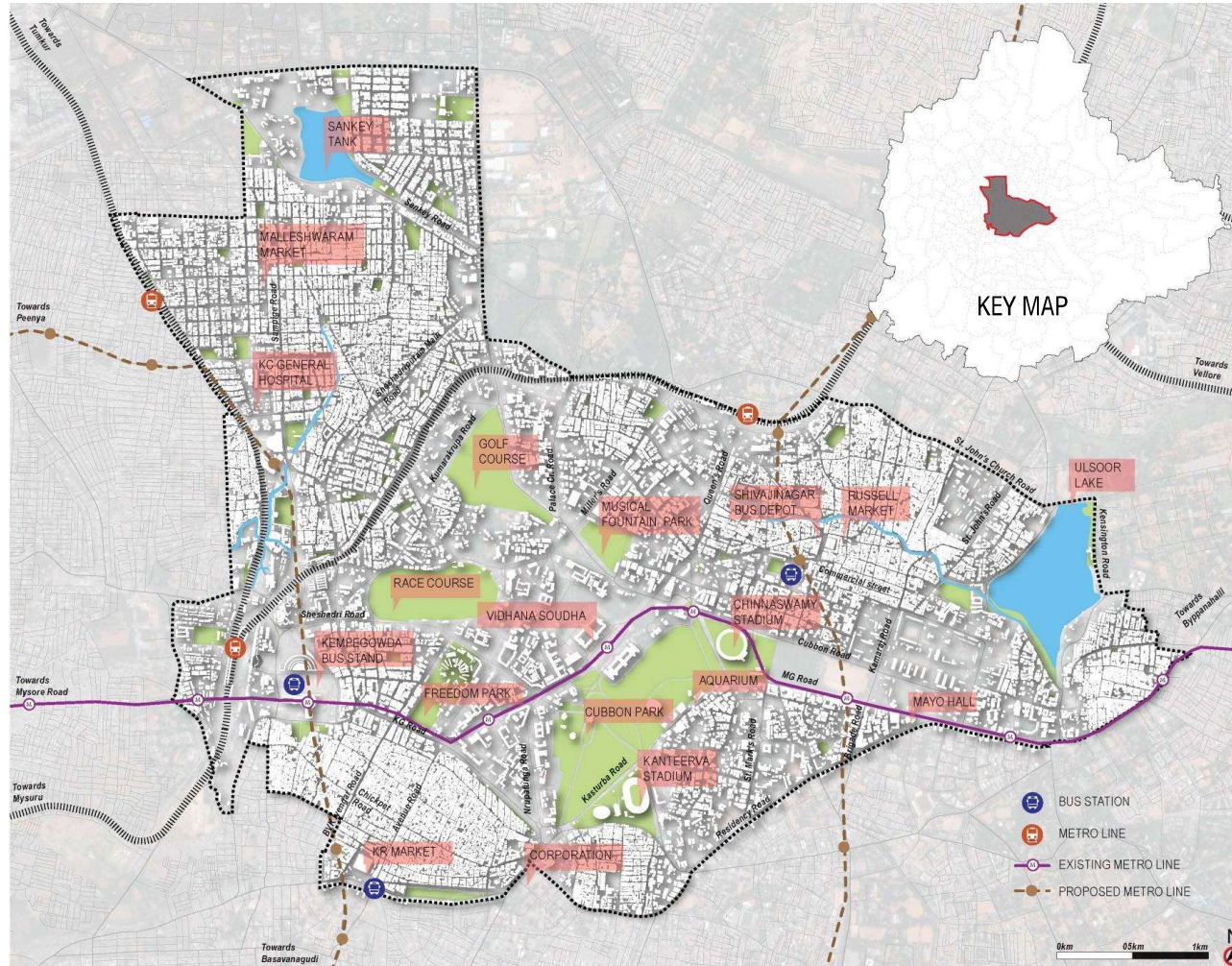
Area for maximum impact

"BELONGS TO, AND IS USED BY EVERYONE"



AREA BASED DEVELOPMENT

Revitalizing the city center of Bengaluru



AREA: 5380 ACRES (21.8 SQ KM)
POPULATION : 5,24,000
17 WARDS

ABD LANDMARKS



CBD & ADMIN

Vidhana Soudha
City corporation
MG road
Kasturba Road



OLD CITY CORE

Historic Pete - Chickpet
Avenue Road
Cantonment
Brigade road
Commercial street



SOCIO-CULTURAL

Kempegowda Museum
Vishveshwariah Tech. Museum
JN Planetarium
Aquarium
Chinnaswamy stadium



PARKS AND LAKES

Cubbon Park
Bal Bhavan
IG Musical fountain
JN Planetarium
Ulsoor Lake
Sankey Tank



CULTURE & HERITAGE

Bangalore Fort
Historic markets
St.Mary's Basilica
Jama Masjid
Temples in Malleshwaram
& Ulsoor



TRANSIT HUBS

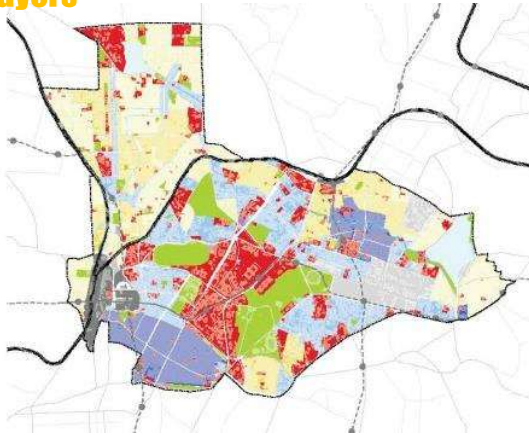
2 city bus stations
2 metro interchanges
and city railway station



SMART BENGALURU

AREA BASED DEVELOPMENT

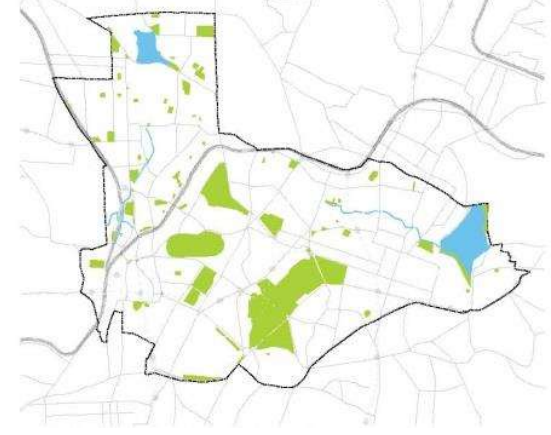
City Layers



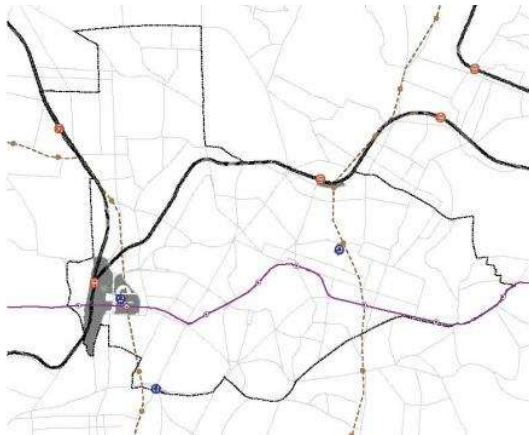
LAND USE MAP
28% Commercial, 19% Public & Semi Public, 13% Mixed Use



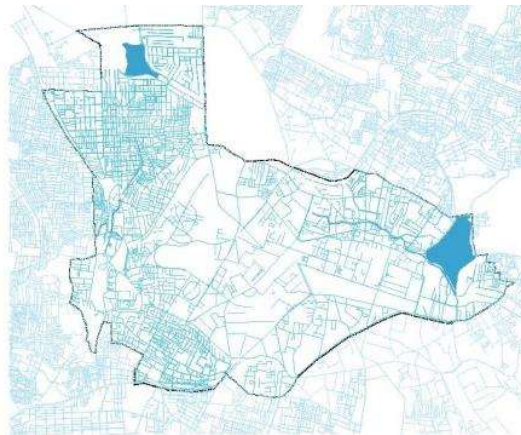
BUILT vs OPEN
29% Built up



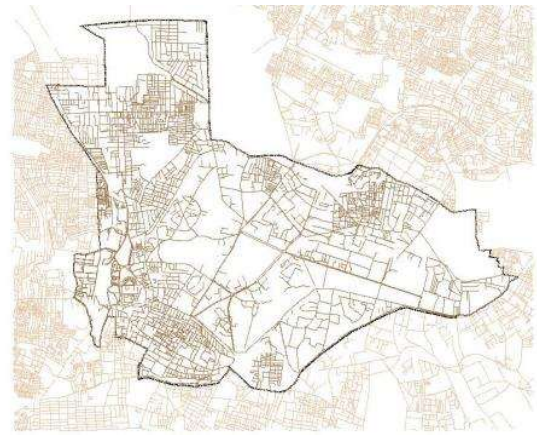
PARKS, OPEN SPACES & LAKES
13% Open space



TRANSPORTATION NETWORK
97 km of major Roads



STORM WATER DRAIN NETWORK
556 km within ABD



SEWERAGE NETWORK
417 km within ABD



AREA BASED DEVELOPMENT

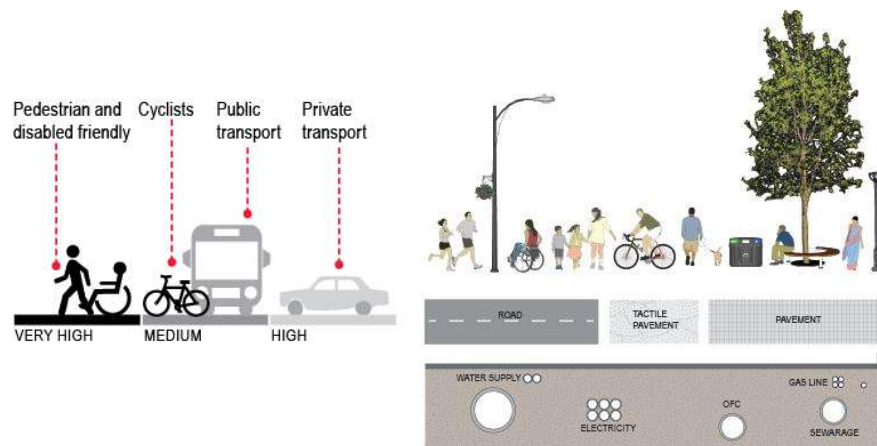


1. Revitalization Of Historic Heart Of City

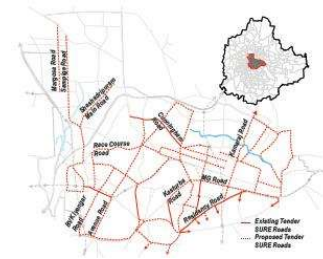
Redevelopment under Tender S.U.R.E.

Proposed for 66% of Major roads in ABD

- **Infrastructure revitalization** (increase in capacity and coverage)- Surface drainage, sewage network, water supply, power and telecommunication
- The city core needs **highest degree of walkability** owing to highest footfall count



“...into a walking, cycling, tree lined, economically vibrant, environmentally sensible and inclusive center”



Tender S.U.R.E.	Promoting NMT	Improved public amenities	Energy efficiency/ conservation	Smart S.U.R.E.
Uniform travel lanes, underground services, smart parking, landscaping, street features	Pedestrian walkway and cycling network, app enabled public bicycle share points	Smart bus shelters with digital information boards, smart dustbins, environmental sensors, e-toilets, water ATMs, smart telecom towers	Vehicle charging points, Solar lighting/ energy efficient lighting, electric buses, e rickshaw stands with charging points	Based on PPP annuity model. Cost recovery on O&M from telecom companies

EXISTING PRECEDENT: 9 Roads – 11.5 km executed; 150 electric buses (BMTc proposal)

AREA BASED DEVELOPMENT

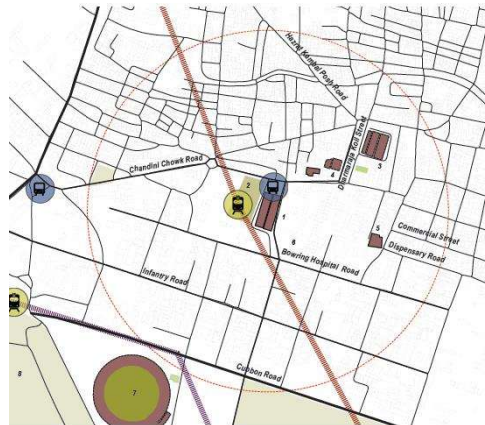


2. Integrated Mobility Towards Creating Vibrant Destination

Shivajinagar Bus Depot + Russell Market, Kempegowda Bus Stand

Integrated Development Of Transport Hub, Market, Civic Park & Connecting Road

- Building on Tender SURE and Proposed Metro Station
- Connecting Shivajinagar bus depot to Russell Market through overhead pedestrian bridge
- Reuse of underutilized public space and infrastructure towards economic viability



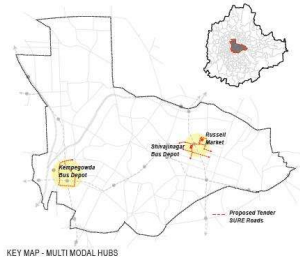
1. Shivajinagar BMTCL Bus Depot
2. Civic park
3. Russell Market
4. St. Mary's Basilica
5. Mosque
6. Browing Hospital
7. Chinnaswamy Stadium
8. Cubbon Park



Shivajinagar Bus Depot Proposed Plan



Shivajinagar Bus Depot - Proposed



Redevelopment of Shivajinagar Bus Depot & Civic Park

Bus routing in and around bus depot. Integrate metro network to bus transportation to increase regional connectivity

Increased space allocation for economic activities

smart cubicles/ start-up hubs/ incubators within CBD, possibly on under utilized public holdings

Integrated development of Russell Market

Retrofitting of historic market building

Improved parking in CBD

App enabled smart car parking with vehicle charging points

Retrofitting of Kempegowda Bus Stand (Majestic)

Digital information boards, way finding features, CCTV surveillance



AREA BASED DEVELOPMENT

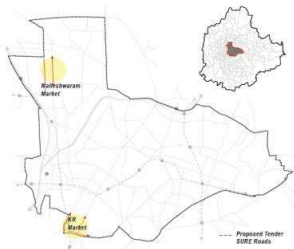


3. Upgradation & Redevelopment Of Historic Economic Centres

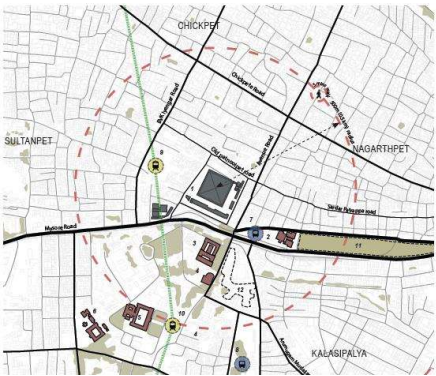
K R Market, Malleshwaram Market

Footfall of 30,000/ day, 1249 vendors

“Towards resilient markets: Connectivity with Metro Station, Reuse of existing unused municipal built assets and Solving the garbage crisis”

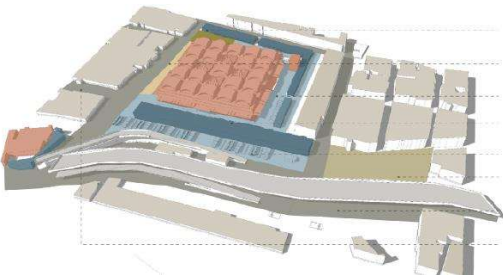


- **Retrofitting of historic market center** through improved public transport connectivity and efficient circulation management
- **Efficient vending and increased retail footprint** through efficient space management
- **Integrating proposed metro station to market** through over head pedestrian bridge



KR Market Context map

- 1. KR Market
- Landmarks
 - 2. Jama Masjid
 - 3. Gavi Dental College
 - 4. Bangalore Fort
 - 5. Victoria Hospital
 - 6. Minis Eye Hospital
- Bus stand
 - 7. K R Market bus stand
 - 8. Kalagipalya Bus Depot
- Metro Station
 - 9. Chikpet Metro Station
 - 10. K R Market Metro Station
- Parks
 - 11. Silver Jubilee Park
 - 12. Open ground



Proposals for market redesign

- Hygienic vending conditions by setting up effective waste management system
- Strengthening existing loading / unloading platforms
- Designated vending areas for better pedestrian movement
- Retaining heritage character of market precinct
- Pedestrian friendly shopping experience
- Enhancing existing transit hub
- Reducing congestion and improving mobility
- Connectivity with metro station



KR Market - Proposed entrance

Redesign of KR Market

Heritage conservation, organizing vehicular and pedestrian movement around the market, provision of loading unloading bays and time managed goods movement

Organized vending

Creating designated vending zones with plazas – smart vendor kiosks

Solid waste management

Sensor based smart dustbins, recycling unit, composting unit

Energy efficiency/ conservation

Roof top solar panels, solar lighting/ energy efficient lighting

Redevelopment of Malleshwaram Market

CCTV surveillance, e-toilets, water ATMs, Public bicycle share points



AREA BASED DEVELOPMENT

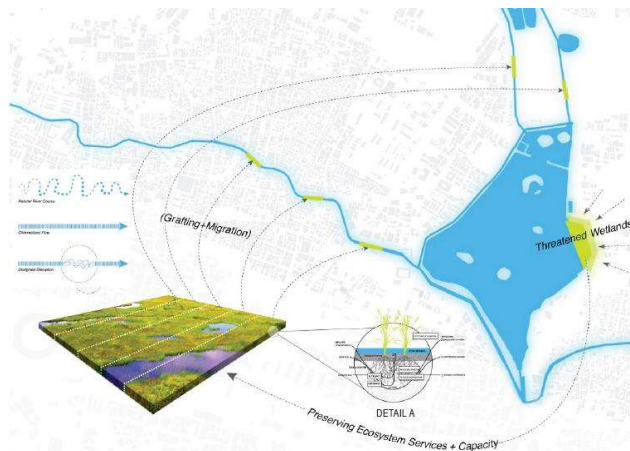


4. Innovation Of Downstream Clean Up Of Drainage System

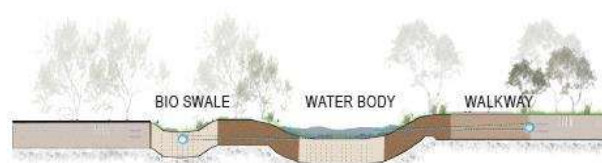
Ulsoor Lake, Sankey Lake

200 acres of lake precincts & 2.5 km of nallah clean up

- **Green initiatives** for water body clean up
- **Increased ground water** recharge
- Creating a **vibrant and accessible public space** within the ABD



Initiatives for Ulsoor lake and nallah clean up

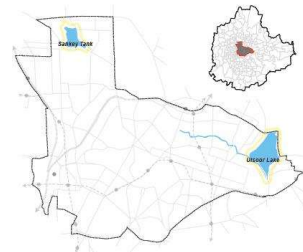


Bio swales along lake and nallah



DEWATs (Decentralised waste water treatment) for reducing contamination of water body

“Developing a water body as a replicable model towards reclaiming the dying historic drainage system of the city”



Proposed pedestrian and bicycle network around Ulsoor lake

Ulsoor lake and nallah clean up	Creating public space	Improved public amenities	Sankey tank
Introduction of DEWAT system, Green Bridges for natural filtration and Bio-Swales in lake and nallah for reducing water contamination	Lake front development as an active public space, Pedestrian and cycle network around the lake	Provision of e-toilets, Smart bins, Public space features/ furniture, public bicycle share points, energy efficient lighting, Solar trees and CCTV installation for improved security	Pedestrian and cycle network around the lake, smart features, children's play equipment, public space furniture



AREA BASED DEVELOPMENT



5. Protection & Redevelopment Of The Centrally Located Parkland

Cubbon Park
25000 foot fall / day (weekends)

“ to become a local community asset ”

- Revitalizing the central city green through improved pedestrian connectivity and place making
- Integrate green network to heritage and cultural landmarks around Cubbon park



Cubbon Park



6. Increasing Affordable Housing Stock Through Slum Redevelopment

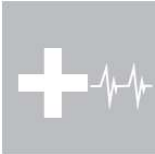
Swathanthra Palya Slum (Gandhinagar)
Housing + road + Utility Services

“ improving quality of life ”

- Upgrading of housing units
- physical infrastructure - roads, water supply, sanitation, street lighting
- social infrastructure - Community centre



Proposed slum upgradation



7. Retrofitting Of A Health Care Facility

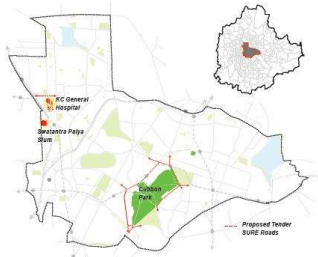
K.C. General Hospital
Building redesign + abutting road under Tender SURE

“ improving a health care centre ”

- Building redesign of health care centre
- Heritage conservation of building



Existing KC General Hospital

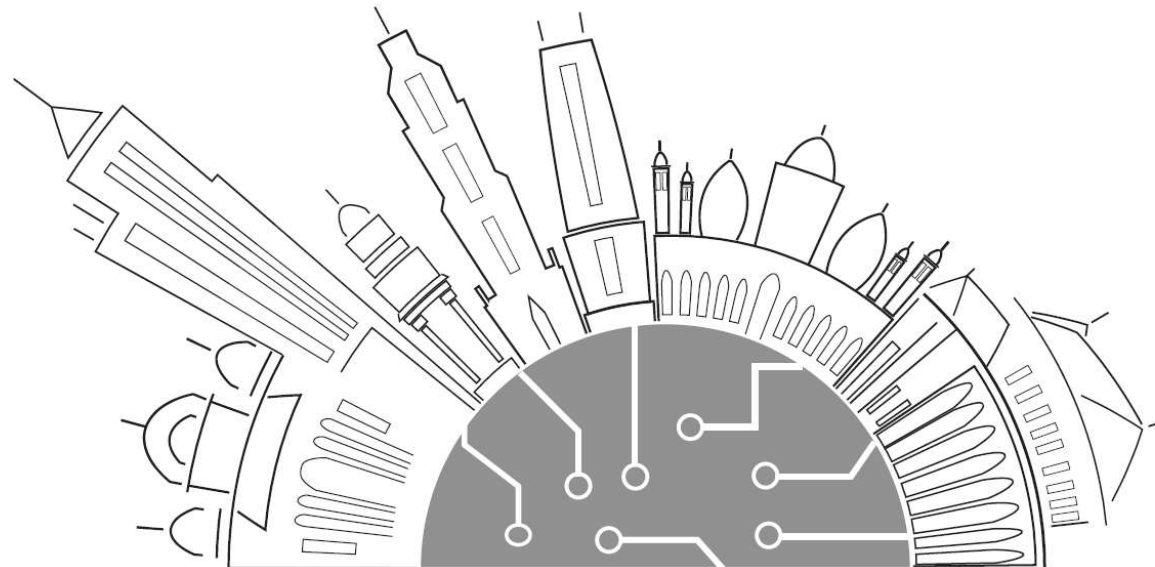


Cubbon Park	Safety and security	Rain water harvesting	Affordable housing	K.C. General hospital
Digital information board (activities, events, eateries and places of interest), tourist information kiosk, seating	CCTV Installation and improved energy efficient lighting	Collection and re use towards ground water recharge	Housing units, physical infrastructure, social infrastructure, roof top solar panel on community centre, energy efficient solar lighting	CCTV surveillance, Roof top solar panels, public bicycle share points, Tender SURE road for abutting



PAN CITY INTERVENTIONS

Project: Citizen Centric E-Governance and Services



PAN CITY PROPOSALS

Project 1: Citizen Centric E-governance And Services



1. Grievance Management

- Central Command Centre for accepting citizen queries across agencies
- Smart Kiosk for people without digital reach

Governance



2. Municipal Finance Reforms

- Municipal Audits
- City dashboard as a single MIS
- Surveying wards and streets to accurately report condition of infrastructure



3. Participatory Budgeting

- Campaigns and drive to involve residents and publish the collated, analysed results
- Citizen app that consolidates neighbourhood and ward level inputs from the citizens



4. Improvement of Neighborhood Safety

- Community Policing
- CCTV monitoring



5. Improvement in Property Tax Collections

- 3D building mapping for city using drone technology for property assessment, public assets, regulation and revenue augmentation



6. B-TRIPS (Bengaluru Transport related information and planning system)

- Single smart card for all public transport
- PIS system



7. Public Project Management

- Online project information system displaying projects information

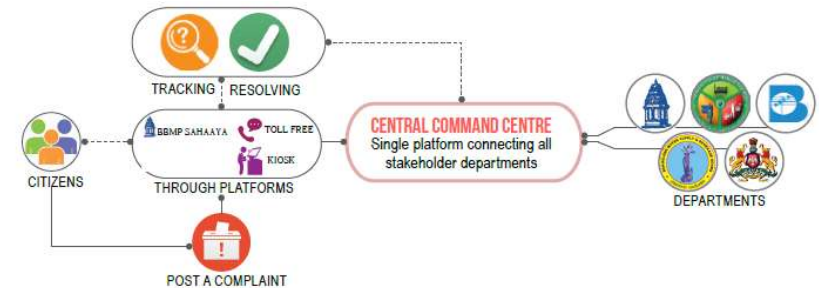
Public Services



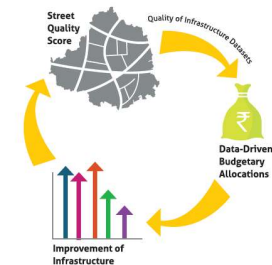
8. Data for Citizens

- Open data portal is based on the ideology of providing spatial data to citizens

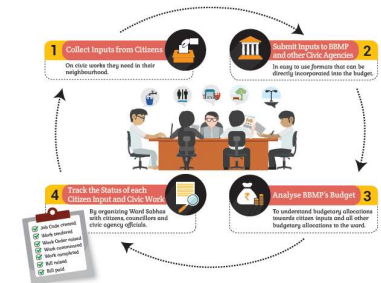
1. Grievance Management: Grievance redressal steps



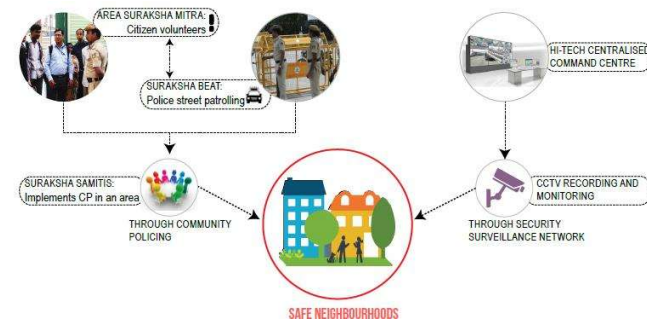
2. Municipal Finance Reforms: Street Quality Report Benefits



3. Participatory Budget



4. Key Initiatives for Safe Neighbourhoods



Thank You



SMART BENGALURU